Water Resources - Rivers

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Introduction

➢ The total area drained by a river and its tributaries is called the drainage basin.

➢ India's drainage systems can be categorized into two primary origins:

Himalayan Rivers and Peninsular Rivers, and also by types:

- Coastal Rivers
- Inland Drainage.

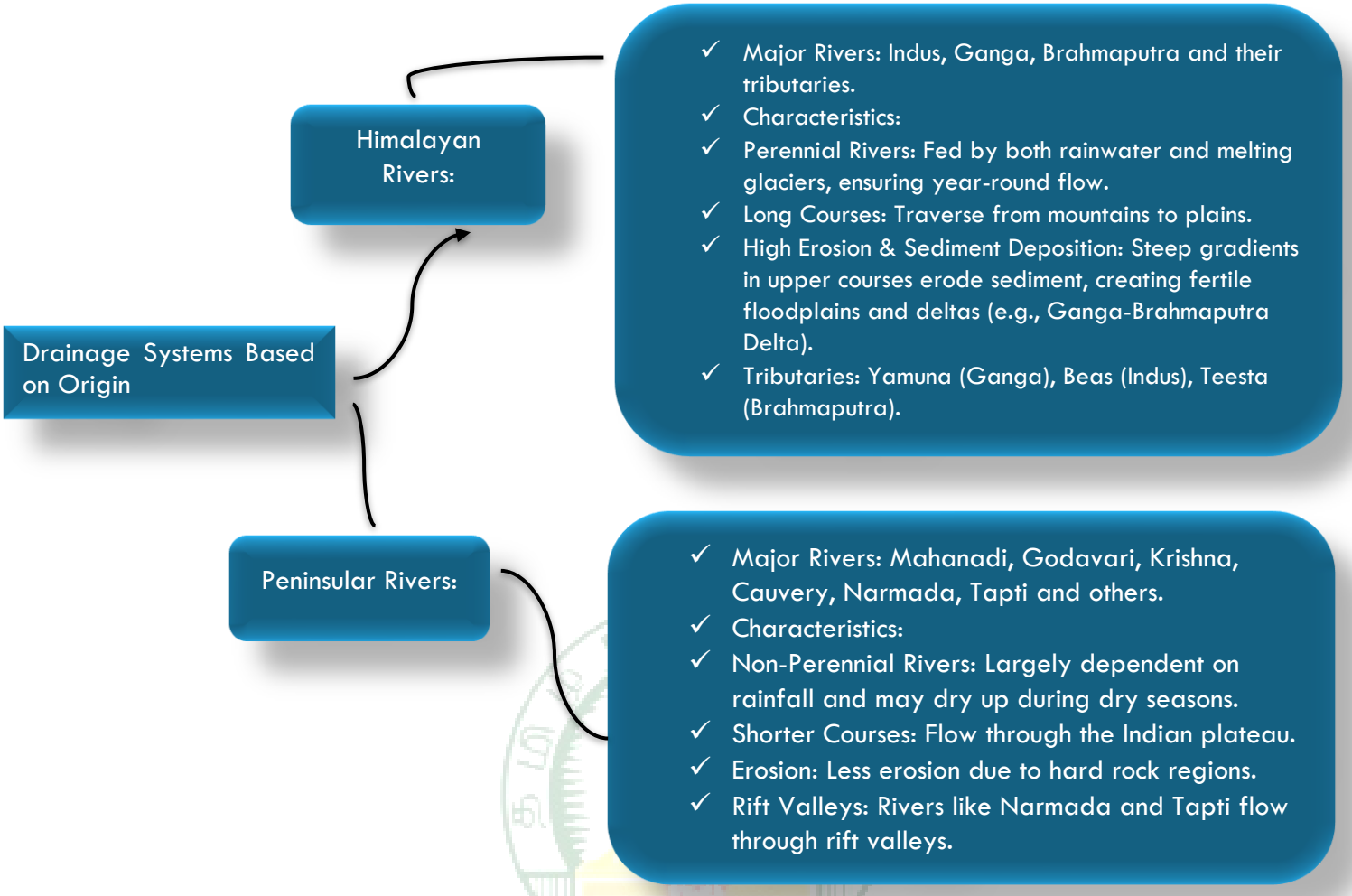
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Drainage Systems Based on Type of Drainage:

Himalayan Rivers, Deccan Rivers, and Coastal Rivers

Himalayan Rivers	• Large perennial river systems like Ganga, Brahmaputra, and Indus.
Deccan Rivers:	• Flow from central and southern Indian plateau, forming deltas (east-flowing rivers) or estuaries (west-flowing rivers like Narmada, Tapti).
Coastal Rivers	• Short rivers draining directly into the sea. Examples include rivers • in Kerala (e.g., Periyar).

Inland Drainage Basin (Endorheic Basin)

Inland Rivers	• Inland Rivers: These do not drain into seas or oceans but into inland basins, salt lakes, or deserts. Example: Sambhar Lake (Rajasthan) and Luni River (Rann of Kutch)

The Bay of Bengal Drainage (East Flowing Rivers):

- Major Rivers: Ganga, Brahmaputra, Mahanadi, Godavari, Krishna, Kaveri, Subarnarekha.
- Flow Direction: Rivers flow from west to east, draining into the Bay of Bengal.
- Features:
  - Long course and large deltas (e.g., Ganga-Brahmaputra Delta, Godavari Delta).
  - Large basin areas.

The Arabian Sea Drainage (West Flowing Rivers):

- Major Rivers: Narmada, Tapti, Sabarmati, Mahi, Periyar.
- Features:
  - Shorter course.
  - Form estuaries (e.g., Narmada, Tapti).
- Flow Direction: Rivers flow from east to west, draining into the Arabian Sea.

Special Case: Narmada and Tapti

- Both flow almost parallel in central India, emptying into the Arabian Sea.
- The Narmada flows through a rift valley between Vindhya and Satpura ranges, and Tapti flows further south, also through a rift valley.

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Himalayan Rivers in Detail:

Indus River
- Source: Northern slope of Kailash range in Tibet near Manasarovar Lake.
- Length: 2,880 km (709 km in India).
- Flows through Jammu & Kashmir, turns south to enter Pakistan.
- Tributaries: Jhelum, Chenab, Ravi, Beas, and Sutlej.
- Crosses Himalayas via a deep gorge (5,181 m).
- Empties into the Arabian Sea south of Karachi

Jhelum River:
- Source: Verinag, southeastern Kashmir.
- Flows through Wular Lake and Pir Panjal Range.
- Joins the Chenab at Trimmu

Chenab River
- Source: Bara Lacha Pass in Zaskar Range.
- Joins Sutlej after receiving waters from Jhelum and Ravi rivers.

Sutlej River:
- Source: Manasarovar-Rakas Lakes in Tibet.
- Flows through Shipki La (Tibet-Himachal Pradesh border), with the Bhakra Dam built on it.
- Forms part of the boundary between India and Pakistan.

Beas River
- Source: Near Rohtang Pass.
- Flows through Dhauladhar Range and joins
- Sutlej at Harike (Punjab).
- Entirely in India.

Ravi River
- Source: Kullu Hills, near Rohtang Pass.
- Crosses the Dhaola Dhar Range and enters Pakistan

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Indus Water Treaty:

- Signed: 19th September 1960 between India and Pakistan.
- Allocation: India is allocated only 20% of the Indus River system's total discharge for its use.

The Ganga River:

- ✓ Largest Drainage System in India.
- ✓ **Origin:** The river originates as Bhagirathi from the Gangotri Glacier in Uttarkashi district, Uttarakhand, at an elevation of 7,010 m.
- ✓ **Length:** 2,525 km.
- ✓ **Tributaries:** ○ From the north: Gomti, Gandak, Kosi, Ghagra. ○ From the south: Yamuna (largest tributary), Son, Chambal.
- ✓ Known as Padma in Bangladesh.
- ✓ Forms the Sundarbans Delta with the Brahmaputra before draining into the Bay of Bengal.

Ganga River Characteristics:

- ✓ At Devprayag, the Alaknanda River joins Bhagirathi to form the Ganga.
- ✓ Pollution: Ranked as the fifth most polluted river in the world (2007).

- ✓ Major Tributaries of Alaknanda:
- ✓ East Trishul at Karnaprayag.
- ✓ Pindar from Nanda Devi.
- ✓ Mandakini/Kali Ganga at RudraPrayag.
- ✓ Dhauliganga.

- ✓ Major Tributaries of Bhagirathi:
- ✓ Bheling.
- ✓ Bishenganga

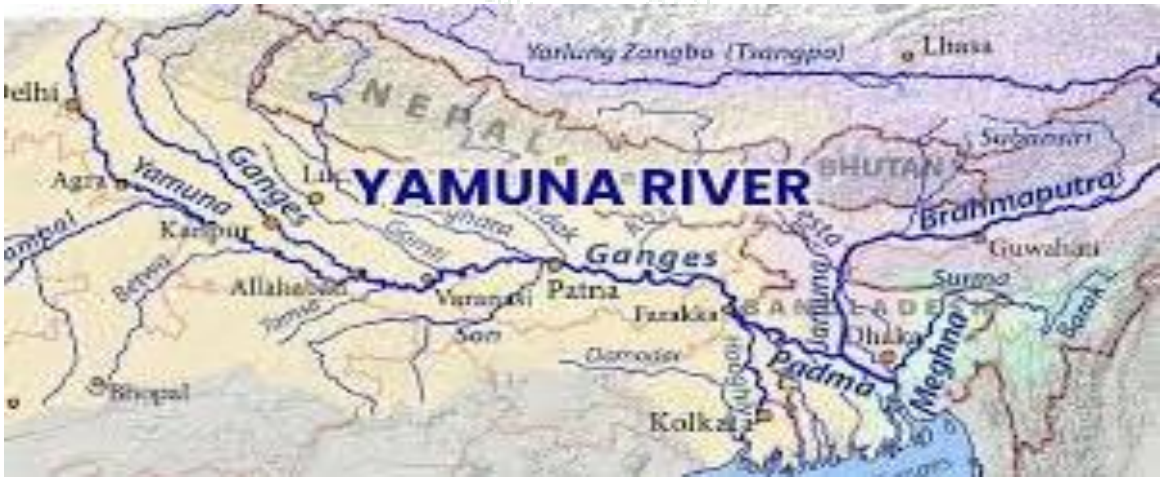
- ✓ The Ganga enters the plains at Haridwar and meets the Yamuna at Prayagraj.
- ✓ At Farakka, it bifurcates into Bhagirathi-Hugli and Padma-Meghna.
- ✓ The Brahmaputra joins the Padma-Meghna at Goalundo.

- ✓ Ganga-Brahmaputra Delta:
- ✓ Largest Delta in the World.
- ✓ Area: 58,752 sq km.
- ✓ Known for frequent flooding and being low-lying, highly indented by coastline

Yamuna River:

- Largest Tributary of the Ganga.
- **Origin**: Yamunotri Glacier on Bandarpunch Peak, Uttarakhand, at an elevation of 6,000 m.
- Flows through Nag Tibba, Mussoorie, and Shiwalik Ranges.
- Joins the Ganga at Triveni Sangam in Prayagraj.
- **Length**: 1,376 km.
- Tons River is a major tributary, joining the Yamuna below Kalsi.

Tributaries:



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Chambal River
- Origin: Janapao Hills, Vindhya Range.
- Join the Yamuna.
- Length: 1,050 km.
- Dams: Gandhi Sagar, Rana Pratap, Jawahar Sagar, and Kota Barrage.
- Key feature: Forms badland topography

Banas River
- A tributary of the Chambal, joins near Sawai Madhopur

Sind River
- Origin: Vidisha Plateau, Madhya Pradesh.
- Joins the Yamuna after flowing for 415 km.

Betwa River
- . Origin: Bhopal district, Vindhya Range.
- Joins the Yamuna near Hamirpur.
- Tributary: Dhasan

Ken River
- Origin: Barner Range, Madhya Pradesh.
- Joins the Yamuna near Chila

Son River
- Origin: Amarkantak Plateau near the Narmada source.
- Joins the Ganga near Danapur, Bihar

Ramganga River
- Origin: Garhwal district, Uttarakhand.
- Joins the Ganga at Kannauj

Ghaghra River
- Source: Gurla Mandhata peak, Tibet.
- Known as Karnaili in Nepal.
- Tributaries: Sarda, Sarayu (Sarju), Rapti

Kali River
- Origin: High glaciers of the trans-Himalaya.
- Forms boundary between Nepal and Kumaon.
- Joins the Ghaghra

Gandak River	•.Origin: Tibet-Nepal border at 7,620 m. •Tributaries: Kali Gandak, Mayangadi, Bari, Trishuli. •Joins the Ganga at Hajipur, Bihar
Burhi Gandak River	•Origin: Western slopes of Sumesar hills. •Joins the Ganga near Monghyr
Kosi River	•Known as Saptkaushiki. •○ Originates in eastern Nepal. •○ Joins the Ganga near Kursela. •○ Forms three main streams: Tumar, Arun, Sun Kosi. •○ Prone to flooding, known as the 'Sorrow of Bihar'.

The Brahmaputra River:

Regions and Names:

- ❖ **Tibet:** Known as Tsangpo (meaning "The Purifier").
- ❖ **China:** Referred to as Yarlung Zangbo and Jiangin.
- ❖ **Assam Valley:** Flows as Dihang or Siong, south of Sadiya, and is known as Brahmaputra.
- ❖ **Bangladesh:** Called the Jamuna River, and when combined with the Ganga, it forms the Padma River. The merged flow later joins Meghna and creates a large estuary before entering the Bay of Bengal.

Origin and Course:

- ❖ The Brahmaputra originates from the Chemayungdung Glacier of the Kailash Range, located east of Lake Mansarovar in Tibet at an elevation of about 5,150 m.
- ❖ Known as Tsangpo in Tibet, it flows for about 2,900 km (with 900 km in India).
- ❖ It enters India through the Dihang Gorge in Arunachal Pradesh.
- ❖ **Tributaries:** ○ Tista, Manas, Barak, and Subansiri.

Course & Characteristics:

- ❖ Begins as Tsangpo in Tibet, receiving tributaries like Raga Tsangpo and Ngangchu.
- ❖ Turns southward around Namcha Barwa (7,756 m), entering India and cutting through the Dihang Gorge to emerge in the Assam Valley near Sadiya.
- ❖ It flows as Siong and then Dihang in Assam before becoming known as the Brahmaputra from Sadiya onward.

- **North Bank Tributaries:** Subansiri, Kameng, Dhansiri, Raidak, Tista (historically a Ganga tributary).
- **South Bank Tributaries:** ○ Dibang, Lohit, Kolong.

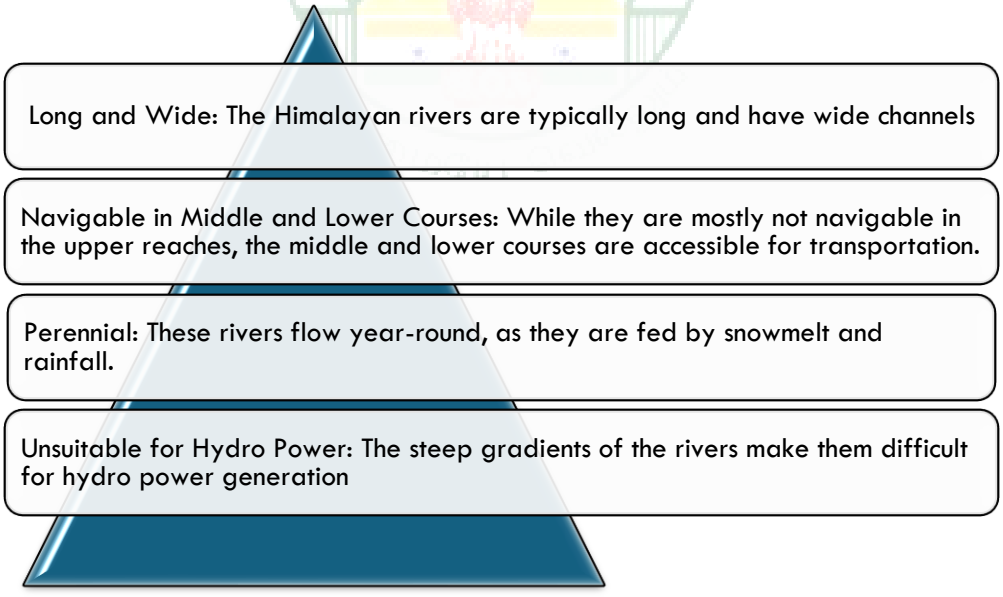
Physical Characteristics:

- **Braided Channel**: The river has a shifting channel due to heavy silt load and meandering patterns.
- **Width:** The river reaches nearly 16 km at Dibrugarh, where it forms several islands, including Majuli, which is the world's largest river island.
- **Flooding**: Known for disastrous floods due to monsoonal rains and heavy sediment flow, earning it the nickname "River of Sorrow."
- **Navigation**: It is navigable for 1,384 km up to Dibrugarh, facilitating inland water transport.

Bangladesh Flow:

- The Brahmaputra enters Bangladesh near Dhubri, flowing as Jamuna for about 270 km.
- It joins the Ganga at Goalundo and the combined flow is referred to as the Padma.
- The river eventually merges with the Meghna and forms a broad estuary before entering the Bay of Bengal.

Charact eristics of Himalayan Rivers

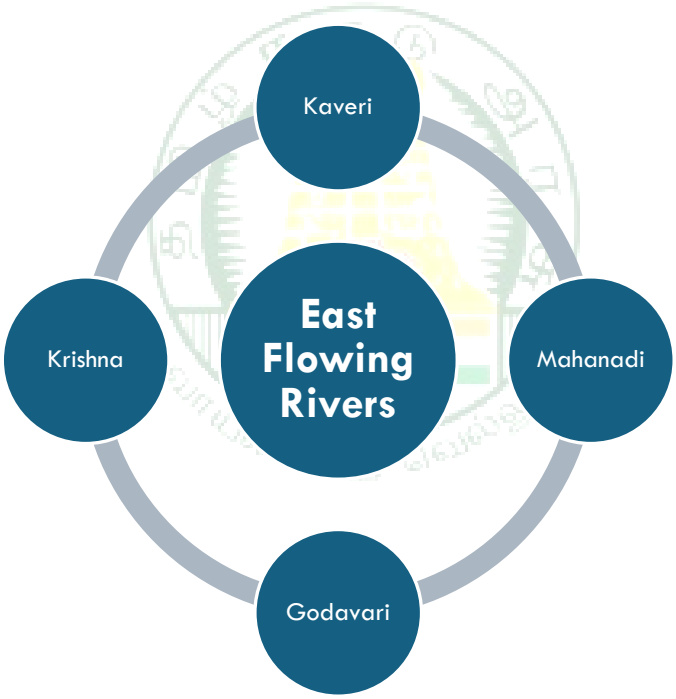


Peninsular Rivers:

Seasonal Rivers: Rivers in the Peninsular India are typically non-perennial and flow mainly during the monsoon season. ● They are divided into

- East Flowing Rivers
- West **Flowing Rivers**

East Flowing Rivers



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Mahanadi

- ✓ **Origin:** Near Sihawa in Raipur district, Chhattisgarh.
- ✓ **Length:** 851 km.
- ✓ **Tributaries:** ○ Seonath, Telen, Sandur, Chitartala, Genguti, Nun.
- ✓ Course: After receiving the Seonath River, it turns east and flows into Odisha,eventually emptying into the Bay of Bengal.
- ✓ **Major Infrastructure:**
  ○ Hirakud Dam: Located at Sambalpur, it's one of India's largest dams, forming a 55 km long lake.
- ✓ Key Urban Centers: Raipur, Durg, Cuttack.
- ✓ **Industries:** Iron & Steel (Bhilai), Aluminum factories (Hirakud, Korba).
- ✓ **Flooding:** The Mahanadi is one of the most active silt-depositing rivers in the Indian subcontinent.

Godavari

- ✓ **Nickname:** "Dakshin Ganga" (Southern Ganga).
- ✓ **Origin:** Trimbakeshwar, Nasik district, Maharashtra.
- ✓ Length: 1,465 km.
- ✓ **Tributaries:** ○ Left Bank: Dharna, Penganga, Wainganga, Wardha, Pranahita, Pench, Kanha, Sabari, Indravati. ○ Right Bank: Pravara, Mula, Manjra, Peddavagu, Maner (Manjra is the longest right bank tributary at 724 km).
- ✓ **Delta Formation:** The river divides into two branches at Rajahmundry: Vashishta Godavari (west) and Gautami Godavari (east), forming one of India's largest deltas before entering the Bay of Bengal.
- ✓ **Lake:** Kolleru Lake is located in the delta region.
- ✓ **Importance:** The Godavari basin covers multiple states, including Maharashtra, Telangana, Andhra Pradesh, Chhattisgarh, and Odisha.

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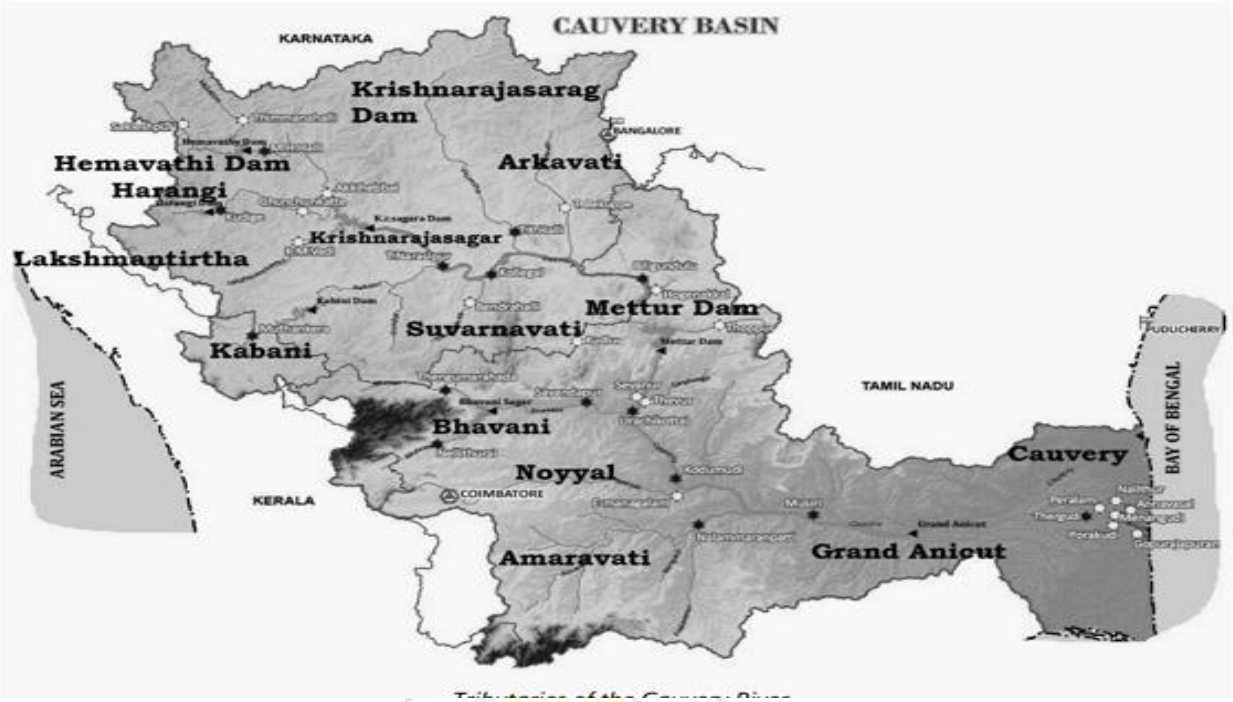
Krishna

- ✓ **Origin**: A spring at Mahabaleshwar, Western Ghats, Maharashtra.
- ✓ **Length**: 1,400 km.
- ✓ **Tributaries**: ○ Right Bank: Ghatprabha, Malprabha, Tungabhadra. ○ Left Bank: Bhima, Musi, Munneru.
- ✓ **Key Features**: ○ Koyna River is a significant tributary, with the Koyna Dam responsible for the 1967 earthquake.
- ✓ **Major Projects**: Tungabhadra, Ghataprabha, Nagarjunasagar, Malaprabha, Bhima, Bhadra, Telugu Ganga.
- ✓ **Hydro Power**: Key stations include Koyna, Tungabhadra, Sri Sailam, Nagarjuna Sagar.
- ✓ **Economic Importance**: Rich in mineral deposits, it holds great potential for industrial development.

Kaveri

- ✓ **Nickname**: **"Dakshin Ganga"** due to its spiritual significance.
- ✓ **Source**: Talakaveri, Brahmagiri Range, Kodagu, Karnataka (elevation: 1,341 m).
- ✓ **Basin Area**: 81,000 sq km (spanning across Tamil Nadu, Karnataka, Kerala, and Puducherry).
- ✓ **Tributaries**: ○ Left Bank: Harangi, Hemavati, Shimsha, Arkavati.
- ✓ **Key Features**:
  ○ Right Bank: Lakshmantirtha, Kabbani, Suvarnavati, Bhavani, Noyil, Amaravati.
  ○ Shivanasamudra Waterfalls (101 m high), utilized for hydroelectric power generation.
  ○ Mekedatu: A wide gorge, also known as "Goats Leap".
  ○ Hogenakkal Falls: The river takes a southerly course and enters the Mettur Reservoir.
- ✓ **Delta Formation**: It splits into two branches near Srirangam Island, forming the Cauvery and Vennar rivers after crossing the Grand Anicut.

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- ✓ Alternative Name: Also known as Uttara Pinakini.
- ✓ Course: Flows eastward, draining into the Bay of Bengal.
- ✓ Total Length: 597 km from origin to outfall.

- ✓ Source: Rises in the Chenna Kesava hills of the Nandidurg range, Chikkaballapura district, Karnataka.

Pennar River

- ✓ Basin Area: Extends over Andhra Pradesh and Karnataka.
- ✓ Tributaries
  - Left Bank: Jayamangali, Kunderu, Sagileru.
  - Right Bank: Chiravati, Papagni, Cheyyeru

- ✓ Basin Boundaries:
  - North: Erramala range.
  - East: Nallamala and Velikonda ranges (Eastern Ghats).
  - South: Nandidurg hills.
  - West: Narrow ridge separating it from the Vedavati valley (Krishna Basin).
- ✓ Other notable ranges: Seshachalam (famous for Red Sanders) and Paliconda ranges.

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Subarnarekha River

- ✓ **Origin**: Ranchi Plateau, Jharkhand.
- ✓ **Course**: Forms the boundary between West Bengal and Odisha in its lower reaches.
- ✓ **Length**: 395 km.
- ✓ **Outflow**: Joins the Bay of Bengal, forming an estuary between the Ganga and Mahanadi deltas.

Brahmani River

- ✓ **Formation**: Created by the confluence of the Koel and Sankh rivers near Rourkela.
- ✓ **Length**: 800 km.
- ✓ **Basin Geography**: ○ North: Bounded by the Chhotanagpur plateau. ○ West and South: Mahanadi basin. ○ East: Bay of Bengal.
- ✓ **Flow**: Passes through Jharkhand, Chhattisgarh, and Odisha, draining into the Bay of Bengal.

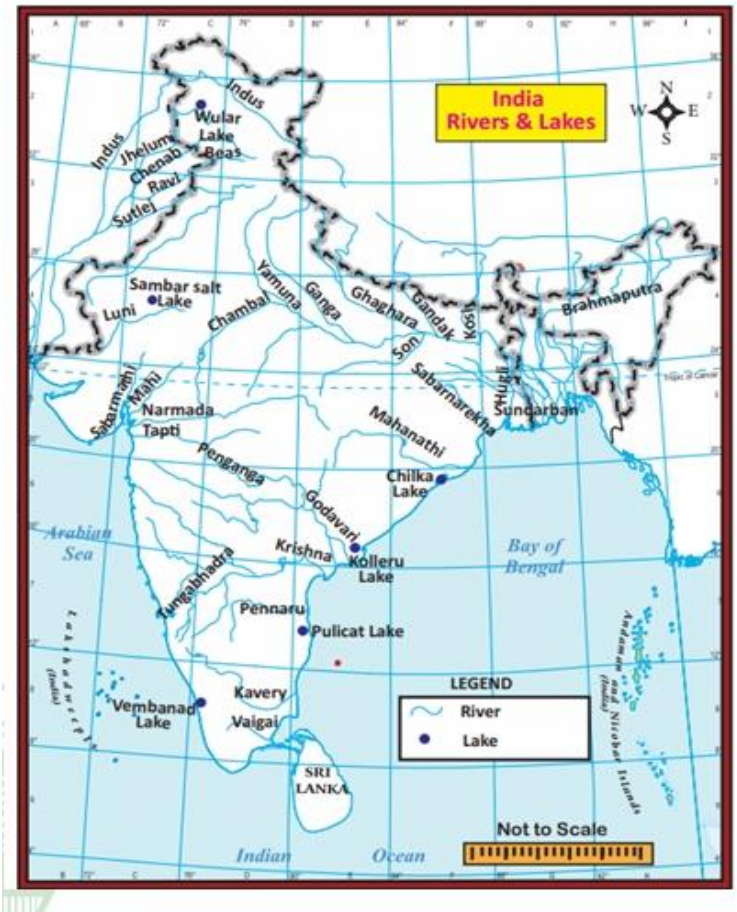
Ponnaiyar River

- ✓ Characteristics: A small stream confined to the coastal area.
- ✓ Regions Covered: Flows through Tamil Nadu, Karnataka, and Andhra Pradesh.
- ✓ Basin Boundaries: ○ North-West and South: Various ranges of the Eastern Ghats (Velikonda, Nagari, Javadu, Shevaroy, Chitteri, Kalrayan). ○ East: Bay of Bengal.

Vaigai River

- ✓ Location: South of the Cauvery delta.
- ✓ ● Significance: The longest stream among several in the region between the Cauvery and Kanyakumari
- ✓ ● Basin Geography:
  - ○ West: Bounded by Varushanadu, Andipatti, Cardaman, and Palani hills.
  - ○ East: Palk Strait and Palk Bay

West Flowing Rivers of Peninsular India:



- ✓ **Fewer and Smaller:** West-flowing rivers are fewer in number and shorter compared to east-flowing rivers.
- ✓ **Flow Mechanism:** These rivers flow through geological faults created by tectonic activity from the formation of the Himalayas.
- ✓ **No Deltas:** Unlike east-flowing rivers, they do not form deltas but estuaries when reaching the Arabian Sea.
- ✓ **Silt & Sediment:** Limited sediment transport due to the rivers flowing over hard rocks, which also restricts delta formation.

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Narmada River

Source	**Amarkantak Plateau, Madhya Pradesh (elevation: 1,057 m).**
Right Bank	Hathni River, Orsang River, Barna River, Kolar River.
Left Bank	Smaller tributaries draining from the northern slopes of the Satpura Range.
Estuary	Forms a 27 km long estuary before outfalling into the Arabian Sea through the Gulf of Cambay.
Significance	The largest among the west-flowing rivers in Peninsular India.
Length	1,312 km. Basin Area: 98,796 sq km

Tapti River

Source	Multai, Betul district, Madhya Pradesh (elevation: 752 m).
Length	724 km. ● Basin Area: 65,145 sq km.
Tributaries:	Right Bank: Suki, Gomai, Arunavati, Aner. Left Bank: Vaghur, Amravati, Buray, Panjhra, Bori, Girna, Purna, Mona, Sipna.
Physical Regions	Hilly regions like the Satpura and Ajanta hills, and fertile plains of Khandesh suitable for agriculture
Outfall	Drains into the Arabian Sea through the Gulf of Cambay.

Sabarmati River:

Origin	Formed by the combined streams of the Sabar and Hathmati rivers, from the Aravalli hills, Udaipur, Rajasthan
Length	371 km. ● Basin Area: Saurashtra and Gujarat
Tributaries	Left Bank: Wakal, Hathmati, Vatrak. ○ Right Bank: Sei.
Basin Boundaries	Bounded by Aravalli hills (north), Rann of Kutch (west), and Gulf of Khambhat (south).
Rainfall	Ranges from 100 cm in the southern part of the basin to minimal rainfall in Saurashtra.

Mahi River:

Origin	**Vindhya Range, Madhya Pradesh (elevation: 500 m).**
Length	583 km. ● Basin Area: Spans across Madhya Pradesh, Rajasthan, and Gujarat.
Geographical Boundaries:	North/North-West: Aravalli hills. ○ East: Malwa Plateau. ○ South: Vindhya Range. ○ West: Drains into the Gulf of Khambhat.
Hydropower Projects	Mahi Bajaj Sagar Dam and Kadana Dam.
Outfall	Flows into the Arabian Sea.

Luni River:

Origin	Western slopes of the Aravalli range, near Ajmer.
Significance	It is the only major river basin in Western Rajasthan, an arid zone.
Course	Flows southwest through Rajasthan, ultimately draining into the Rann of Kutch where it disappears in the marshes.
Physical Characteristics	The river tends to widen due to erodible banks and a sandy riverbed.
Length	511 km.

South Indian Rivers Characteristics:

- ✓ Origins: These rivers primarily originate from the Western Ghats.
- ✓ Nature: Short and narrow, typically non-perennial.
- ✓ Usage: Excellent for hydroelectric power generation but not suitable for navigation

Central Government Initiatives in River Conservation

Namami Gange

Objective: This flagship program is dedicated to the comprehensive conservation of the Ganga River, addressing multiple aspects like Pollution control , Sewage treatment , Riverfront development

Focus Areas: Cleaning the river, creating sustainable sewage treatment plants, and developing riverfronts along the Ganga for cultural, recreational, and ecological purposes.

National River Conservation Plan (NRCP)

Objective

- NRCP aims to improve the water quality of major rivers in India by focusing on:
- Sewage treatment plants
- Pollution control measures
- Community participation
- Environmental sustainability

Focus Areas

- Cleaning the river, creating sustainable sewage treatment plants, and developing riverfronts along the Ganga for cultural, recreational, and ecological purposes

River Basin Management Plans

Objective: Developed for major river basins, these plans focus on integrated and sustainable management of water resources, addressing concerns such as:

- Water availability
- Water quality
- Ecosystem health

Goal: Ensuring the effective management of river systems, enabling conservation and sustainable usage of water across river basins.

Clean Ganga Fund

Objective	Objective: A special fund set up to support projects focused on cleaning and rejuvenating the Ganga River.
Use	The fund is dedicated to initiatives that enhance water quality and bring about effective solutions for pollution control.

Sujalam Sufalam Yojana

Objective: This scheme is focused on the improvement of water resource management, which includes:

- Construction of water bodies
- Irrigation infrastructure
- Maintenance of water systems for better usage and sustainability.

Other Initiatives

- ✓ **Amrit Sarovar Mission:** Focuses on conservation and rejuvenation of water bodies across India.
- ✓ **Small River Rejuvenation & Conservation Plan:** This is a localized initiative to conserve and rejuvenate small and lesser-known rivers.
- ✓ **Mission Water Conservation:** Encourages water conservation efforts across various regions.
- ✓ **Mass Awareness Programs:** These initiatives help create public awareness about the importance of river conservation.
- ✓ **National Water Awards:** Awards given to recognize excellence in water conservation efforts.
- ✓ **Weekly Water Heroes Contest:** A community-driven initiative aimed at encouraging individuals and groups to take action for water conservation.